

Decentralized Observation: Unanimity, Deadlock, and the Value of Communication

Edition 2, revised after a joint audit

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Abstract

Institutions observe different indicators, and their votes on a shared instrument count only in agreement; the obstruction calculus has so far spoken with one regulator's voice. This paper, the D13 item of the generalization catalogue, models two agencies with split observation rights (agency 1 reads z_1 , agency 2 reads z_2) and a unanimity command protocol: the instrument executes only on agreement, and disagreement deadlocks the system. On the two-patch working system (Abaee, 2026, An obstruction calculus, Section 8): the coordinator (reading the joint state) sustains 28 of 36 safe pairs; the decentralized unanimity regime, enumerated exhaustively over all 256 agency-law pairs ($256 = 16 \times 16$: each agency's law maps its coordinate's four values to two votes), sustains exactly 12 — and exactly 16 coordinated-viable pairs are lost, including the exact instance $\{(1, 2), (3, 1)\}$, viable under each agency alone and under the coordinator but non-viable under *every* unanimity law pair. The audited belief $\{(1, 2), (2, 1)\}$ survives decentralization only through the anti-symmetric voting discipline (one agency's low-stock vote paired with the other's high-stock vote; 16 of the 256 law pairs sustain it — a numerical coincidence with the 16 lost pairs, no identity), and the authority restriction (forbidding agency 2's vote) destroys the entire decentralized kernel ($12 \rightarrow 0$, strict instance: the audited belief). The obstruction is robust to a one-step delayed own reading (agency 2 voting on its previous coordinate value). Communication has exact, computable value: the deadlock shadow (8/8 check families).

Introduction and relation to the programme

The catalogue proposes multiple regulators with split observation rights $y_i = O_i(x)$ and split command rights $u_i \in U_i$, joint action constrained by common knowledge, delays, authority, and enforcement — and warns that it is a larger extension to be added only where institutional constraints are central. The governance focus of the programme is exactly that case; the paper stays minimal — two agencies, one instrument, unanimity — and computes

exactly.

Split rights under unanimity

The realized instrument is the agreed vote: $u = u_1(z_1)$ when $u_1(z_1) = u_2(z_2)$; otherwise the protocol deadlocks — $u = 0$ executes no instrument, and the protocol's declared uncoordinated-extraction rule applies, both coordinates losing one ($z^+ = (\max(0, z_1 - 1), \max(0, z_2 - 1))$); a declared rule of this protocol, distinct from the directed damage 2 of an executed instrument). The laws: agency i 's law maps its coordinate's four values $z_i \in \{0, 1, 2, 3\}$ to votes $\{1, 2\}$, so each agency fields $2^4 = 16$ laws and the enumeration covers $16 \times 16 = 256$ law pairs. The induced closed loop is deterministic for a fixed law pair (u_1, u_2) ; the belief set is updated *pointwise* under the realized instrument (both realizations of the reading tracked), so cycle detection on belief sets is exact. Decentralizability of a belief is an *existence certificate* over law pairs — an institutional-design statement (which beliefs the protocol can sustain), not a belief-indexed controller: no agency needs knowledge of any audited belief; the enumeration certifies what the protocol can and cannot sustain.

Proposition 1 (The deadlock obstruction and its exact boundary). The coordinator's kernel holds 28 of 36 safe pairs; the decentralized kernel holds exactly 12; exactly 16 coordinated-viable pairs are not decentralizable, and one such instance is exhibited — the pair $\{(1, 2), (3, 1)\}$, viable under each agency alone (the z_1 -only and z_2 -only structures of Abaee, 2026, An obstruction calculus, Section 8's structure audit) and under the coordinator, non-viable under every unanimity law pair. The audited belief $\{(1, 2), (2, 1)\}$ is decentralizable only through voting laws including the anti-symmetric discipline — agency 1 voting 1 on $z_1 \leq 1$ and 2 on $z_1 \geq 2$, agency 2 voting 2 unless $z_2 = 2$ — with 16 of the 256 law pairs sustaining it: institutional coordination is load-bearing exactly on the deadlock shadow.

Proof. Exhaustive enumeration over the 256 law pairs with exact trajectory simulation; the exhibited instance and the anti-symmetric law's trajectory are verified directly. The counting identity $28 - 12 = 16$ is the communication value on this family. $\square$

Proposition 2 (Authority and delay). Removing agency 2’s right to vote 2 collapses the decentralized kernel from 12 to 0 (strict instance: the audited belief), and granting agency 2 a one-step delayed own reading does not restore the deadlock boundary: the decentralized kernel under the delayed reading remains strictly below the coordinator’s. (The delay is local memory — agency 2 voting on its own previous coordinate value — not common knowledge of the joint state; the terminology of the first edition is here corrected.) Relation to the lattice companion: its institutional class (aggregate reading with a no-repeat codex) sustains 24 of 36 on the same grid — a different institutional restriction with a different count; no ordering between the unanimity protocol’s 12 and the codex chain’s 24 is claimed.

Proof. Both computations are exhaustive and exact: the restricted law space is the 16 pairs with $u_2 \equiv 1$; the delayed regime lets agency 2 vote on its own previous coordinate value (initialized at the current value) and simulates the augmented (state, previous- z_2) trajectories exactly. $\square$

Verified record (8/8 exact check families)

E1 the coordinator’s kernel (28, the full-observation kernel of Abaee, 2026, An obstruction calculus, Section 8); E2 the decentralized kernel (12) by exhaustion over 256 law pairs; E3 the exact boundary (16 lost pairs), the exhibited instance $\{(1, 2), (3, 1)\}$ (individually sufficient under each agency alone and under the coordinator, jointly non-viable), and the audited belief’s anti-symmetric survival; E4 the setwise communication value ($12 \subset 28$); E5 the authority collapse ($12 \rightarrow 0$, strict instance recorded); E6 the delayed-own-reading robustness of the deadlock boundary; E7 the law-space identity ($256 = 2^4 \times 2^4$) with the deadlock transition as declared ($u = 0$: both coordinates lose one) and the loss instance’s three-way viability pinned; E8 the two sixteens distinguished (16 lost pairs against 16 sustaining law pairs — a coincidence of this grid, no identity).

What is not claimed

Two agencies, one instrument, unanimity: no n -agency theory, no sequential or weighted voting, no mechanism design, no enforcement or compliance dynamics, no negotiation models. The deadlock protocol is the simplest agreement discipline; procedural alternatives (default rules, tie-breakers) are not classified, and the deadlock extraction rule is a declared modelling choice of this protocol, not a consequence of the dynamics. Institutional effectiveness is not estimated from data. The catalogue’s larger-programme warning is adopted: the paper claims the exact deadlock phenomenon and its boundary on the audited system, nothing broader.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were

used. **Code availability.** The verification script `institutional_observation_v2_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all eight check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review of the first edition; the joint audit of the first edition and this revision were prepared with further AI assistance, under the programme’s verification discipline.

References

Abaee, A.: An obstruction calculus for viability under incomplete observation. Submitted. <https://github.com/MIKEAA2020/general-sustainability> (2026)